Daniel Lief

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Education

University of California, Berkeley Berkeley, CA
B.S. Electrical Engineering and Computer Science (EECS) *Expected May 2028*

- **GPA:** 4.0/4.0
- **Honors:** EECS Honors Program (Computational Design)
- **Coursework:** Data Structures and Algorithms, Efficient Algorithms, Machine Learning, Artificial Intelligence, Computer Architecture, Databases and SQL, Computer Security, Data Science
- **Teaching:** Academic Intern for CS 61B (Data Structures & Algorithms), a 1,300-student course; taught weekly discussion sections (25 students each) and held office hours
- **Activities:** Eta Kappa Nu EECS Honors Society, Code For Good Berkeley

Experience

Amazon Sunnyvale, CA
Software Development Engineer Intern *May 2026 – Aug 2026*

- Owned the end-to-end development of a budgeting app tracking $191M+ across 63 Amazon device programs and 6 planning cycles, fully replacing a $30K/year 3rd-party legacy tool with 59 internal users
- Built an API gateway and backend of 20 AWS Lambda endpoints with authorization and Protobuf structure
- Cut average home-page load time by 89% from 14s to 1.5s via parallelized data loading and caching
- Designed efficient AWS DynamoDB schemas, reducing live budget recomputation from 1s to sub-millisecond
- Built AI features on AWS Bedrock, allowing users to edit an entire budget instantly, import any file structure into a proper budget, and use their local agent to get budgeting data through an MCP server
- Extended a separate data analytics AI agent to support file attachments end-to-end using Bedrock AgentCore to build Converse API blocks for the model to parse, and S3 + DynamoDB as storage infrastructure
- Worked around model input limits by implementing an image downscaling algorithm with WebP conversion to preserve quality within size caps, and by building a PPTX converter that emits XML with slide structure

REscan Menlo Park, CA
Software Engineer Intern *May 2025 – Mar 2026*

- Scaled 3D model viewing mobile app by building a map view and onboarding process using Unity and C#
- Developed a web-based 3D model visualization platform adopted by clients including the US Air Force, supporting 5 mesh formats and large point-cloud datasets with real-time rendering controls
- Implemented performance-optimized rendering features including point-budget presets, downsampling, caching, and LODs using React Three Fiber, keeping 3GB+ point-cloud models at interactive framerates
- Shipped interactive UI tools including annotation modes, triangle painting, and per-model transforms
- Integrated AWS Amplify user authentication and S3-backed model storage managing 1TB+ of files

Projects

Engram Github
Cal Hacks AI Hackathon 2026 *Jun 2026*

- Co-built (team of 2) a drop-in MCP proxy that speculatively pre-executes an AI agent's read-only tool calls while it reasons, committing or squashing results like CPU branch prediction (3x speedup in testing)
- Architected the proxy core in Python: JSON-RPC 2.0 / MCP stdio framing, a subprocess client with a demultiplexing reader thread for concurrent dispatch, and a speculation cache to store predicted calls

Skills

Programming: Python, Java, JavaScript/TypeScript, C, C#, SQL

Technologies: React, Next.js, Node.js, Tailwind, AWS (Bedrock, AgentCore, Lambda, DynamoDB, S3, Amplify), Unity, Scikit, XGBoost, TensorFlow, Git